GE Medical Systems

Technical Publications

Direction 2100625

Revision 1

Signa© Advantage™ 1.0T Breast Coil

Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, nature, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785-5052/8*323-5052 immediately after damage is found. At this time be ready to describe name of carrier, delivery date, consignee name, freight or express bill number, item damaged, and extent of damage.

Complete instructions regarding claim procedures are found in section "S" of the Policy & Procedure Bulletins.

3/12/92

GE MEDICAL SYSTEMS

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LIST OF EFFECTIVE PAGES

REVISION	DATE	PRIMARY REASON FOR CHANGE
REV 0	11/15/93	Initial Release
REV 1	6/17/96	Add 60cm values to Table 2-2

PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV
Title Page	1						
Damage	-						
A	1						
B	Blank						
i	0						
ii	Blank						
1-1 — 1-5	0						
1-6	Blank						
2-1	1						
2-2	0						
3-1 — 3-7	0						
3-8	Blank						
4-1 — 4-4	0						
5-1 — 5-3	0						
5-4	Blank						
Data1	0						
Data2	Blank						
Data3 — Data4	0						
	Blank						
GE Logo Pg.	-						

Table of Contents

SECTION 1 - INTRODUCTION

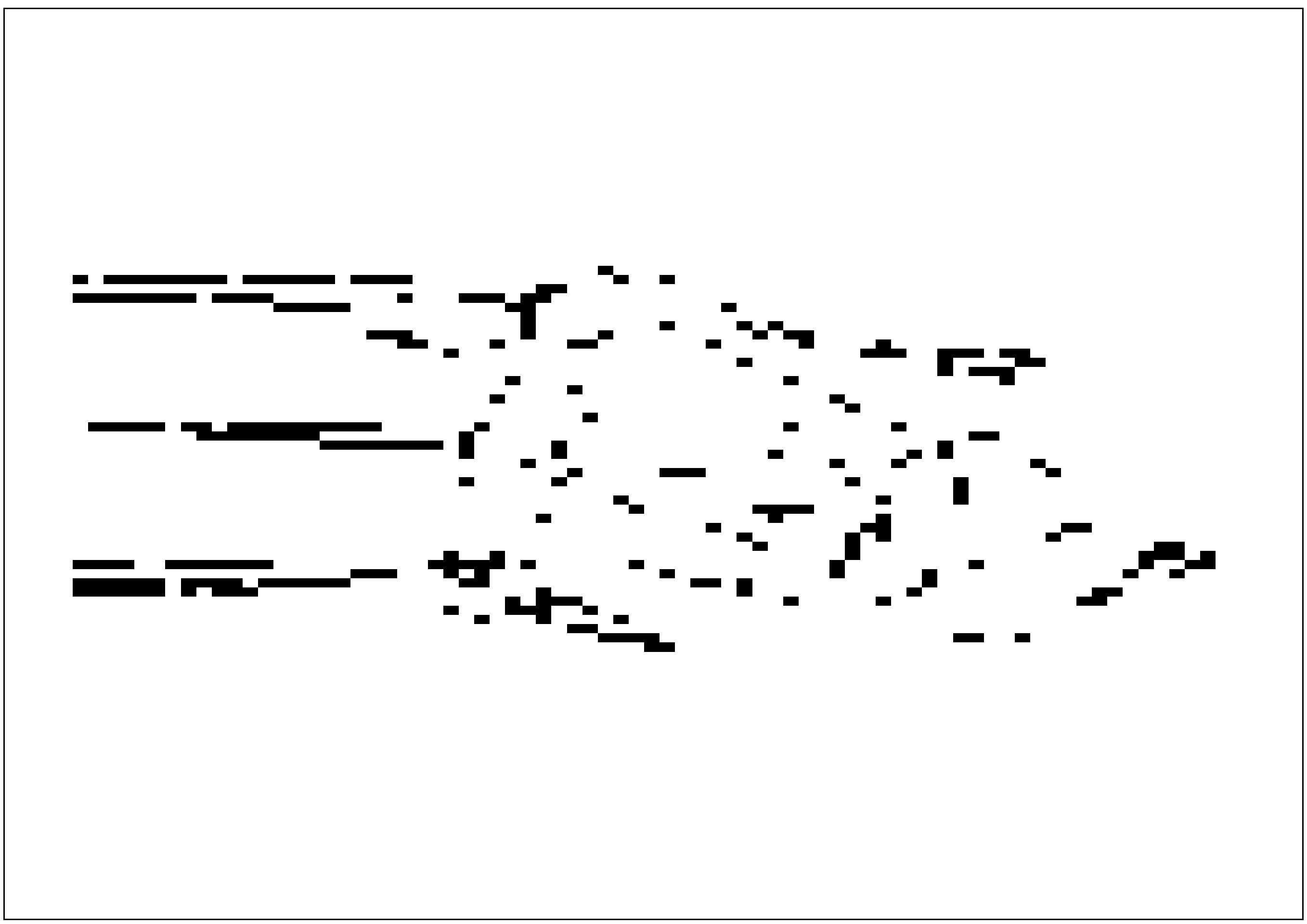
1-1 HOW THE BREAST COIL OPERATES

The Breast coil is a two element receive only antenna array which receives the RF signal from the body coil and sends it to the preamplifier. The coil is used to image the female breast. The coil has two main applications: detection of breast pathology—breast cancer, examination of silicon breast implants.

The Breast coil can be used in two configurations—single-receiver or phased array, determined solely by the external cable used. The coil is shipped with the single-receiver cable installed. A switch on the single-receiver cable allows the Breast coil to be used in three modes: unilateral left, bilateral, or unilateral right. A phased array cable is also shipped with each coil. If the coil is intended to be used as a phased array coil, or if the site purchases the phased array option in the future, the single-receiver coil cable can be removed and the phased array cable installed. Do not discard the extra cable.

The two small individual antenna loop elements provide excellent SNR across a 32cm FOV. The Breast coil is tuned to the 1.0T MR frequency, 42.68 MHz.

The Coil consists of three major assemblies, See ILLUSTRATION 1-1:



BREAST COIL WITH CABLES
ILLUSTRATION 1-1

1-1 HOW THE BREAST COIL OPERATES - (continued)

The two coil element assemblies each contain a printed kapton conductor mated with a circuit board containing the matching and decoupling components. A tuning capacitor is included at the coil interface printed circuit board to enable the coil to be adjusted to exact resonance at 42.68 MHz. The combined series value of the tuning, decoupling, and drive capacitors on the interface printed circuit board was selected to resonate the coil at 42.68 MHz. The ratio of decoupling and tuning capacitor was selected of provide a matched output impedance of 50 ohms under nominal loading conditions. The drive capacitor is composed to two parallel capacitors to provide increased RF voltage rating without resorting to custom high-voltage RF capacitor. See ILLUSTRATION 1-2.

An inductor in series with the coil output allows preamplifier activated multicoil decoupling to be accomplished by the low shunt impedance of the preamplifier input. A PIN diode is installed at this location to aid in transmit decoupling, provide a proper load for the T/R bias driver, and protect the preamplifier. The inductor is adjusted to resonate with the drive capacitor at 42.68 MHz, thus inserting a high impedance in series with the coil loop at the transmit RF field frequency whenever the PIN diode is biased into the **ON** state. Similarly, a low impedance at the coil output, from the input of the PA preamplifiers, will cause one element of the coil system to decouple from the other to provide coil element-to-coil element isolation in PA mode. Because of the coil element positions dictated by the anatomy, no significant isolation from overlap of the two coil elements was possible. The output of the coil is terminated in an SMA connector to enable the output cable to be readily changed.

A passive decoupling circuit, consisting of four crossed diodes and a series inductor is shunted across the decoupling capacitor. The inductor is tuned by resonate with the decoupling capacitor at 42.68 MHz when the crossed switching diodes are forward biased by induced RF power from the transmit excitation field. The added passive decoupling circuit is used to ensure a significant margin of safety in achieving sufficient transmit decoupling. With the large size of the capacitor required at the drive point, the decoupling impedance of the network is not sufficient to provide the desired safety margin.

To allow the preamplifier input impedance to isolate one coil from the other, the electrical length from the shunt PIN diode on the output of the coil electronics to the preamplifier input must be zero, or a multiple of 1/2 wavelength. At 1.0T, the cable would have to be almost 90 inches long. To maintain the cable length used at 1.5T, a PI section network with a impedance of 50 ohms and electrical length of 60 degrees is used at the output of each coil, located on the interface pcb.

The cable set for the PA mode consists of a dual cable having two SMA terminations at the coil end, and a 30 pin connector block designed to mate with the PA coil port on the host system. The output of the left coil element is connected to PA coil port 2 - corresponding to system receiver 1. The output of the right coil element is directed to PA coil port 5 - corresponding to system receiver 0. The physical length of the cable has been extended to 180 degrees by the Pi section network on the coil element to the input of the PA preamplifier. This ensures that the low input impedance of the PA preamplifier provided some isolation between the two coil elements via a decoupling action.

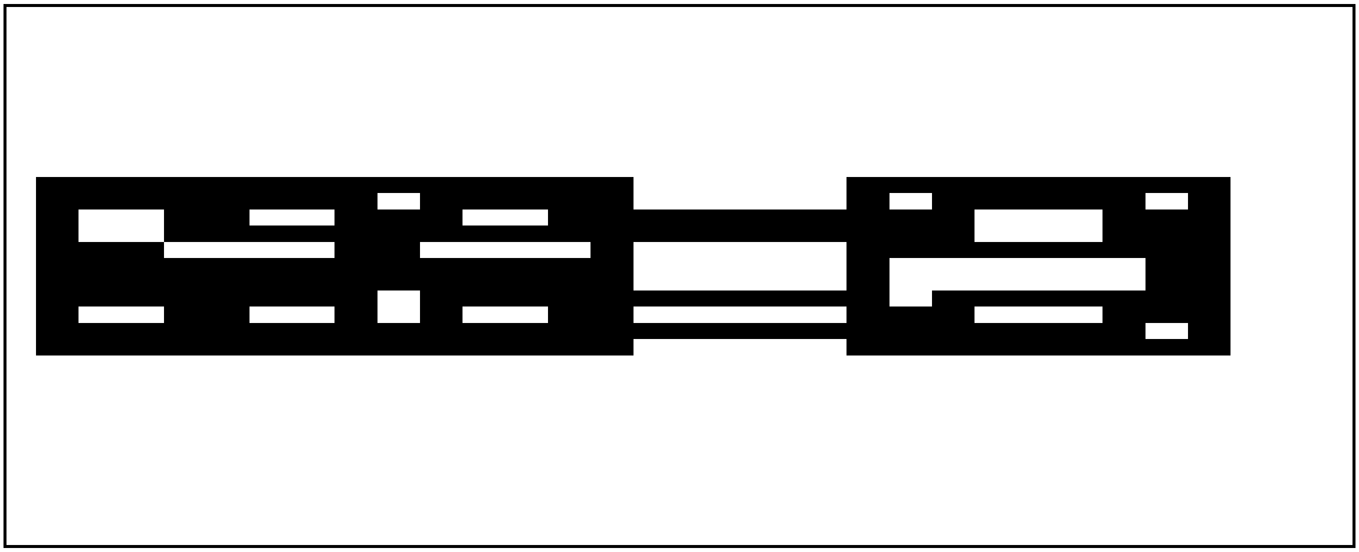
The cable set for the non-PA mode consists of dual cable having two SMA terminations at the coil end, a switch to deselect one coil element when a unilateral scan having optimum signal to noise ratio is desired, a combiner to allow bilateral signal acquisitions, and a 20 pin connector to mate with the host system head/surface coil port.

1-1 HOW THE BREAST COIL OPERATES - (continued)

The combiner consists of two 90 degree, 70.7 ohm Pi sections, one having a positive phase shift, the other having a negative phase shift. As the output components from the two desired networks are of equal reactance but of opposite sign, they are eliminated from the final design. When combined, the signals from the two coil elements are added with a net phase shift of 180 degrees. This is necessary because the two coil elements are identical in design and configuration, but assembled with one coil element rotated 180 degrees relative to the other. As the 50 ohm impedance of each coil element is seen by the system preamplifier through a 70.7 ohm, 90 degree network, the pre amplifier sees two 100 ohm sources in parallel, resulting in the desired 50 ohm source impedance.

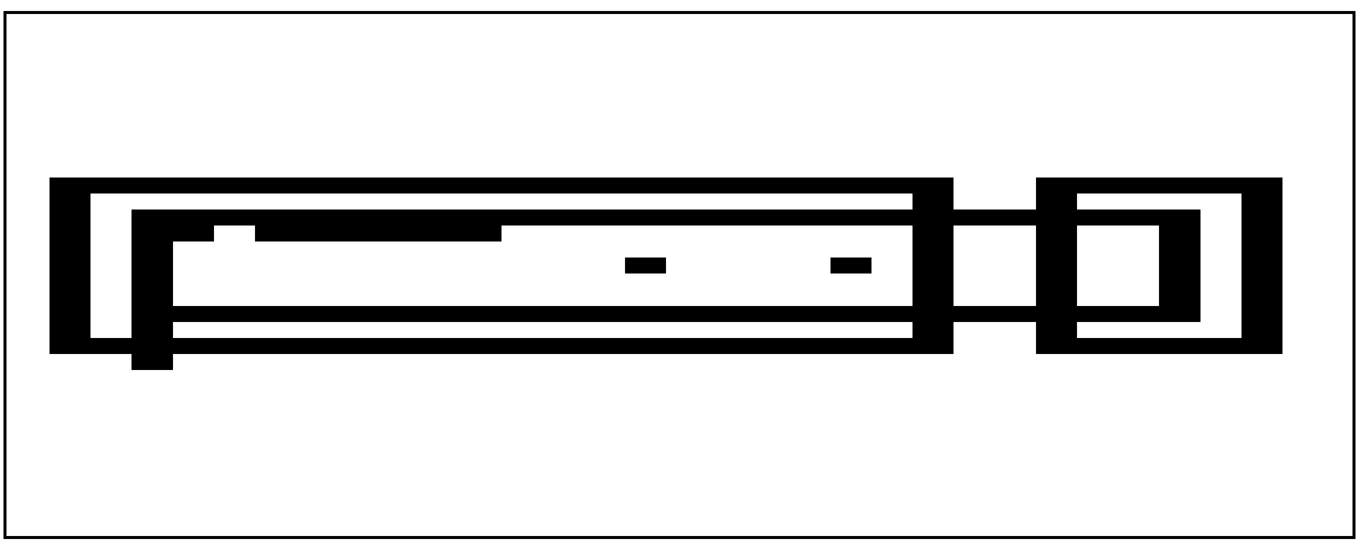
When a unilateral acquisition is desired, the selector switch is used to short out the output from the undesired coil. This has two effects. First, as the cable from the coil element to the switch, in combination with the Pi network on the coil interface card is 180 degrees long electrically, the short created by the switch has an effect similar to biasing on the PIN diode of the unused coil element, thus decoupling it from the desired coil during data acquisition. As the short is an RF short only, the DC bias for the PIN diode is still supplied as under normal imaging. Thus the transmit mode decoupling is unaffected. Second, the switch shorts the unused input to the combiner, causing the Pi section to invert the short into an open at the preamplifier port end of the network. The signal from the desired coil passes through a single Pi section, resulting in a minor mismatch, but in no other loss of signal.

The PIN diode bias for the left-side coil element is obtained from the system receive port bias driver via the series inductor in the 90 degree combiner network. To ensure the optimum level of transmit field decoupling, the bias for the right-side coil element is obtained from the transmit port bias driver of the system head/surface coil port.



BLOCK DIAGRAM FOR SINGLE-RECEIVER CONFIGURATION
ILLUSTRATION 1-2

There are two circuit boards and two antennas in each Breast Coil. See ILLUSTRATION 1-3 for a simple block diagram.



BLOCK DIAGRAM FOR SINGLE COIL
ILLUSTRATION 1-3

1-2 COMPATIBILITY

The Breast Coil is compatible with the following hardware configurations:

- Signa Advantage 1.0T Systems

1-3 RELATED DOCUMENTS

- Direction 15400, Signa Advantage 1.5T, 1.0T, & 0.5T System
- Direction 15400, Signa Advantage 1.5T, 1.0T, & 0.5T Renewal Parts

1-4 ORGANIZATION OF THIS DOCUMENT

This document is divided into the following sections:

Section 1, INTRODUCTION, describes how the Breast coil operates and when and where the Breast coil can be used.

Section 2, SET UP AND CALIBRATION, describes installation procedures.

Section 3, FUNCTIONAL CHECKS, describes the normal power-up sequence.

Section 4, REPLACEMENT/MAINTENANCE, describes field maintenance procedures.

Section 5, RENEWAL PARTS, lists field replaceable parts.

SNR Data Table

Coil Return Form

TLT Table

1-5 ENVIRONMENTAL REQUIREMENTS

Operate and store the Breast coil in the Scanner Room.

SECTION 2 - SET UP AND CALIBRATION

2-1 CHECKING THE SHIPPING LIST

Table 2-1 lists the M1885BR Signa Breast Coil parts. Check that all parts have been shipped.

TABLE 2-1
COIL PARTS

QTY.	ITEM	PART NO.
1	Breast Coil	46-328296P1
1	Phased Array Cable Kit	2100937-5
1	Head Port Cable Kit	2100937-6
2	Arm Pad	46-320757P1
1	Chest Pad	46-320758P1
2	A-sized 3" filler Pad	46-320759P1
2	B-sized 2" filler Pad	46-320760P1
2	C-sized 1" filler Pad	46-320761P1
1	Abdomen Pad	46-320762P1
1	Foot Pad	46-320763P1
1	Face Pad	46-320764P1
2	Cradle Attachment Strap	46-320766P1
5	Disposable Liner, Chest	46-320812P1
5	Disposable Liner, Face	46-320813P1
1	Service Manual	2100625

2-2 INSTALLING THE BREAST COIL

At the terminal, install a new soft key. This key will be used by the operator to select Breast imaging. Name the key "**BREAST**" for single-receiver coil or "**BREASTPA**", for phased array coil. Some proprietary software tools key from the coil name. Use the correct coil name to ensure that these tools function.

Refer to the Signa Advantage 1.5T, 1.0T, & 0.5T System Manual for information on installing soft keys (use Coil default values in Table 2-2)

TABLE 2-2
COIL VALUES

SYSTEM	COIL NAME	COIL TYPE	EXTREMITY COIL	CABLE LOSS	COIL LOSS	RECON SCALE FACTOR	XMIT COIL TYPE	XMIT ATTN	MULTI-COIL.
1.0T	*** See Note	surface	no	1.24	0.103 - 55cm 0.549 - 60cm	Head Coil Recon Scale Fctr.x 0.50	Quad	0	*** See Note

NOTE: For Single-Receiver use **BREAST** and **NO**
For Phased Array use **BREASTPA** and **YES**

2-2 INSTALLING THE BREAST COIL - continued

The Breast Coil is delivered with the single-receive cable installed. Refer to Section 4-2, REPLACING THE EXTERNAL CABLE, to install the Phased Array Cable. (2100937-5).

TABLE 2-3
PHASED ARRAY VALUES

Multi-Coil NAME	# OF REC.	START REC.	STOP REC.	PORT ENABLE MASK	ERROR ENABLE MASK
BREASTPA	2	0	1	5	5

2-3 FUNCTIONAL CHECKS

1. Perform a Body coil scanner performance verification. Refer to Section 3-1, SIGNA ADVANTAGE 1.0T SCANNER VERIFICATION.

2. Perform a Breast Coil performance verification. Refer to Section 3-2, BREAST COIL IMAGING PERFORMANCE VERIFICATION.

3. Perform an SNR Image analysis. Refer to Section 3-3, SNR IMAGE ANALYSIS.

2-4 PERIODIC QUALITY ASSURANCE CHECK

On a periodic basis, such as during planned maintenance, perform the quality assurance check outlined below to ensure that the Breast Coil is operating properly with no appreciable degradation of image quality.

1. Check the external cable for cracks or breaks once each week.

2. Perform a Breast Coil performance verification. Refer to Section 3-2, BREAST COIL IMAGING PERFORMANCE VERIFICATION.

3. Record the date and value calculated in SNR QA DATA TABLE.

4. As instructed in the Data Table, divide the SNR value obtained in the periodic QA check by the original SNR value.

5. If this ratio is not greater than 85%, then there may be a problem with the Breast Coil or the MR system. Contact the GE Service Representative.

SECTION 3 - FUNCTIONAL CHECKS

3-1 SIGNA ADVANTAGE™ 1.0T SCANNER VERIFICATION

Note

An alternative proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" located in Direction15491, *Signa Advantage 1.5T, 1.0T, and 0.5T System Troubleshooting Guide.*

1. Scanner performance in the body coil mode should be verified. The RF power level calibration from the body coil mode is used to verify the surface coil decoupling. Landmark the 100 mm TLT phantom without the coil. Place the TLT Phantom on the Universal Phantom Holder. See ILLUSTRATION 3-1.

2. Set up the scan using the parameters in TABLE 3-1. Coil Type: [**Body Coil**] and Number of Echos: [**4**] should be inserted. Perform an **Auto Prescan** to calibrate the RF power level. Record the value for TG, R1, and R2.

3. Run the scan. Observe the resulting images. Ensure that there are no artifacts of any sort on any of the images.

4. Select [**Cancel**], [**New Series**], to perform SNR comparisons. Acquire two identical images using the previously loaded scan parameters in TABLE 3-1 after inserting Number of Echos: [**1**]. Immediately after acquiring the

TABLE 3-1
SCAN PARAMETERS

ID:	**geservice**	Auto CF:	**[Peak]**
Name:		Receive Bandwidth:	**[16kHz]**
Weight (lb):	**111**	Field of View:	**[48 cm]**
Patient Entry:	**[Head First]**	Slice Thickness:	**[10 mm]**
Patient Position:	**[Supine]**	Interscan Spacing:	**[0]**
Axial/Sag Landmark:	**[Sternal Notch]**	Start Location (I/S):	**0**
Coil Type:	**[++] See Note**	End Location (I/S):	**0**
Scan Plane:	**[Axial]**	No of Scan Locations:	**1**
Image mode:	**[2D]**	FOV Center (L/R):	**0 (P/A): 0**
	[Monitor SAR]	Acq. Matrix: (freq.)	**[256]**
Pulse Sequence:	**[Spin Echo]**	Acq. Matrix: (phase)	**[128]**
Imaging Options:	**[None]**	Frequency Direction:	**[R/L]**
Number of Echos:	**[*] See Note**	Phase FOV:	**default**
Echo Time (TE):	**[20 ms]**	Imaging Time:	**[1 Nex]**
Rep Time (TR):	**[Other] 600 ms**	Contrast:	**[No]**
		Table Delta	**0**

NOTE: * **[4]** for Section 3-1-2; 3-2-2
[1] for Section 3-1-4; 3-2-4
++ **[Body]** for Section 3-1-2; 3-2-2
[Other][Breast] or **[Breastpa]**for Section 3-1-4; 3-2-4

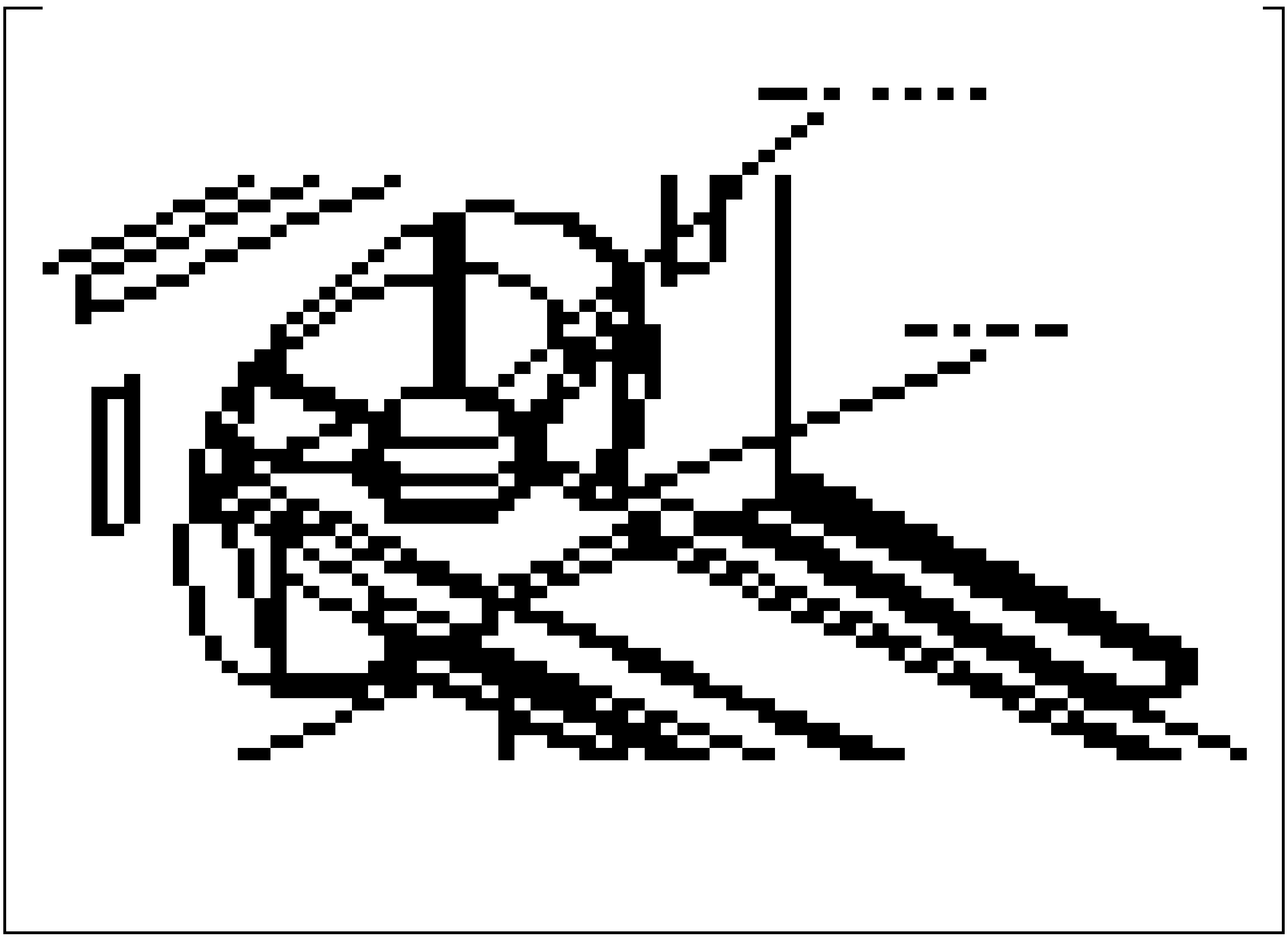
first image, select **Scan** to obtain a second. Note the series number for future reference.

3-2 BREAST COIL IMAGING PERFORMANCE VERIFICATION

1. Set up the Breast coil and 100 mm TLT phantoms as shown in ILLUSTRATION 3-2. A second 100 mm TLT Phantom, P/N 46-317586G1, can be obtained through the GE Service representative.Connect the surface coil cable to the **HEAD COIL PORT (single receiver)** or **PA PORT (PA coil)**. On the non-phased-array configuration, set the **QUICK DISCONNECT BOX** switch to **BOTH**.

2. Select [**Cancel**], [**New Series**], and repeat the scan using the parameters in TABLE 3-1. Coil Type: [**Other**][**Breast**] or [**Breastpa**] and Number of Echos: [**4**] should be used. Use **Auto Prescan** to set the values of R1 and R2 to those in Section 3. If the value of TG does not match the value obtained in step 3-1-2, use **Manual Prescan** to set the value of TG.

3. Run the scan. Ensure that there are no artifacts of any sort on any of the images. Study the image using a **Window Width** and **Window Level** to obtain good contrast. A properly functioning coil will produce an image with a smooth signal pattern, gradually dropping in intensity as the distance from the coil increases, see ILLUSTRATION 3-3. A decoupling failure will cause holes, distortions, or striped patterns in the image. Refer



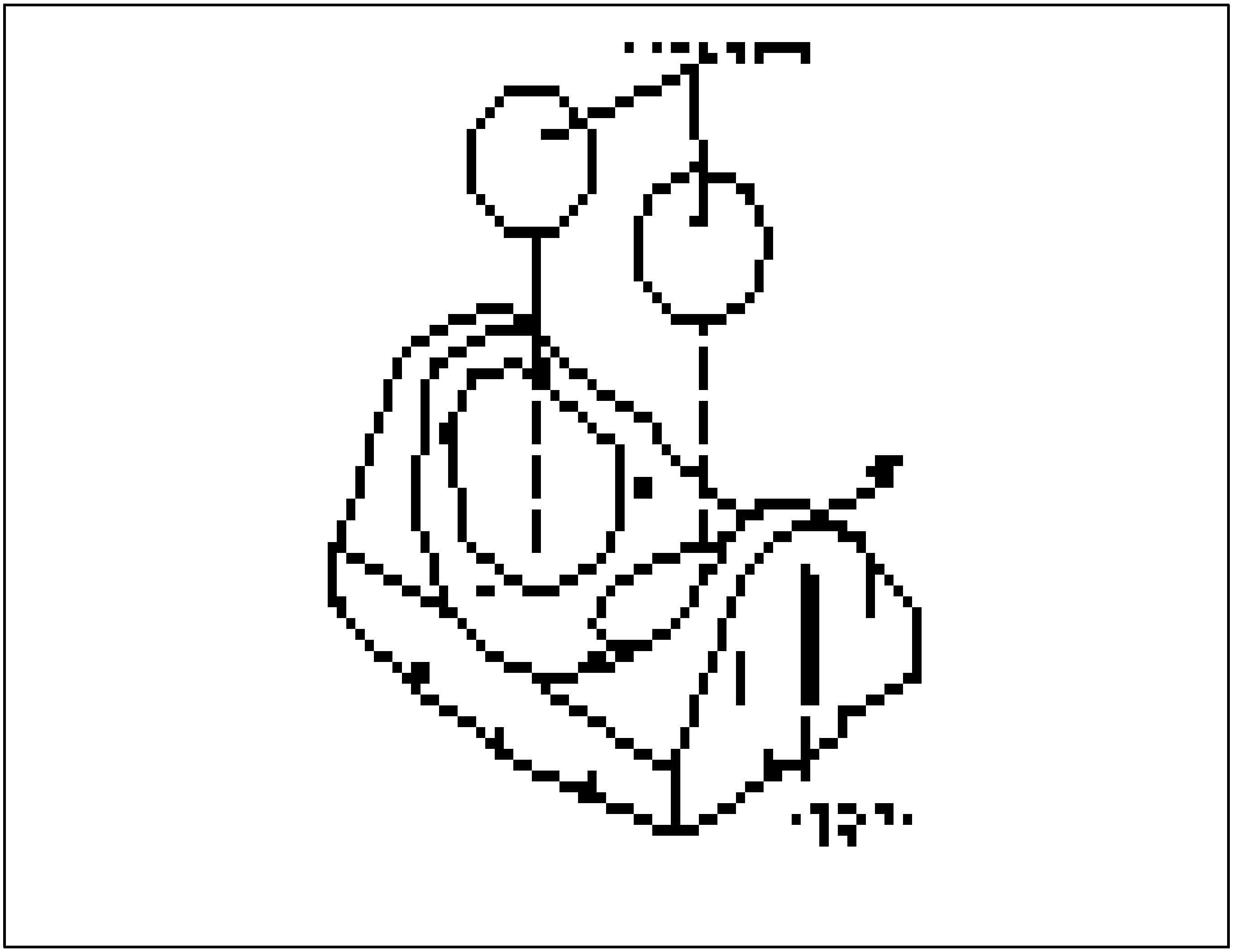
TLT PHANTOM PLACEMENT
ILLUSTRATION 3-1

to Section 3-4 to 3-5 to troubleshoot decoupling failures.

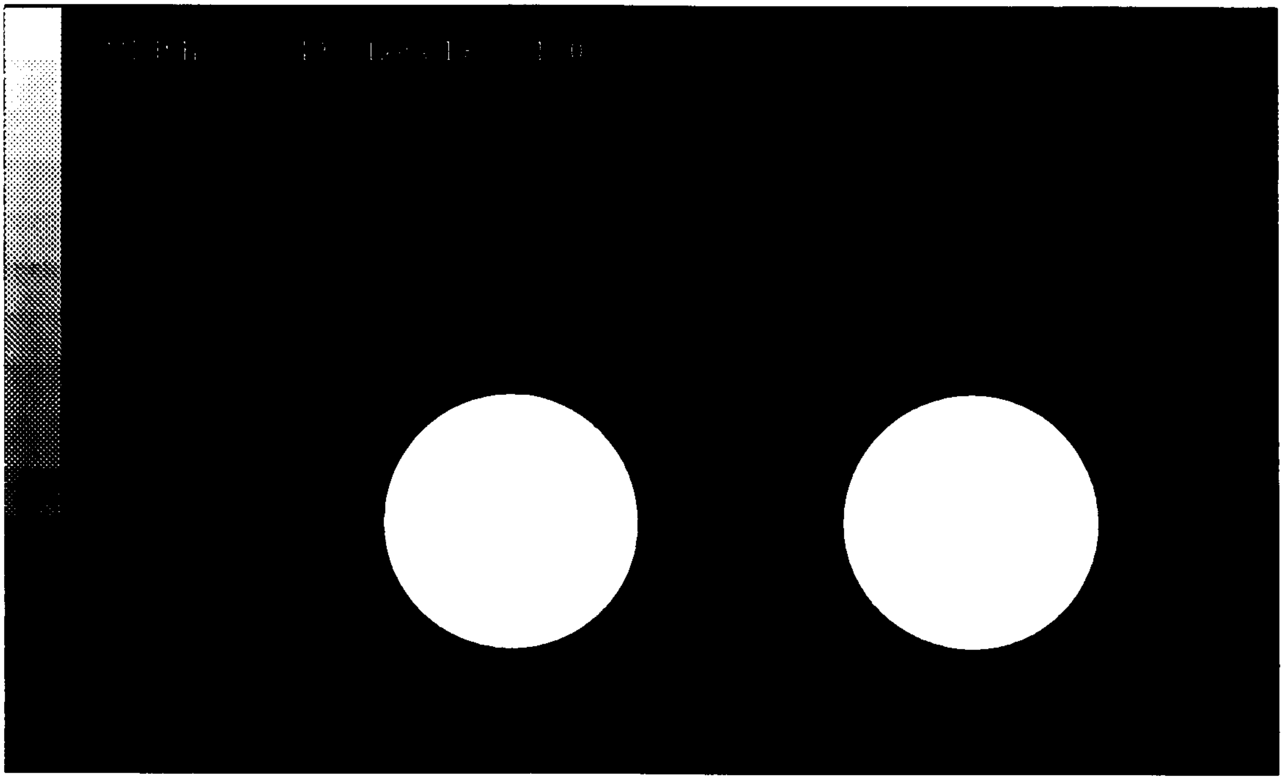
4. Repeat step 2 with Number of Echos: [**1**] inserted. Run the scan. Immediately after acquiring the first image, select **Scan** to obtain a second.

5. Compute the SNR using the procedure described in Section 3-3, SNR IMAGE ANALYSIS. Compare the signal to noise ratio of the first echo of the Breast Coil images with the signal to noise ratio of the first echo of the body coil images of step 3-1-4. The Breast Coil scans should display a signal to noise ratio greater than 2.5 times that of the body coil scans.

6. Inspect the images for visible ghosting (similar in appearance to motion artifact in the phase encoding direction.



TLT PHANTOM PLACEMENT IN BREAST COIL
ILLUSTRATION 3-2



NORMAL DECOUPLING
ILLUSTRATION 3-3

If images contain ghosting artifacts, suspect an intermittent cable or connection. Refer to Section 3-4 to 3-5 for troubleshooting information.

3-3 SNR IMAGE ANALYSIS

Description

The SNR tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center for positioning the ROI. A difference image is created by subtracting the second image from the first and to calculate noise from the subtracted image. Signal value, noise value, and SNR are reported. The difference image can be saved with the results annotated (option).

Procedure

1. Touch [**Utilities**], [**MR Tools**], [**Image Quality**], then [**SNR Test**]. The SNR Test screen is displayed. See ILLUSTRATION 3-4.

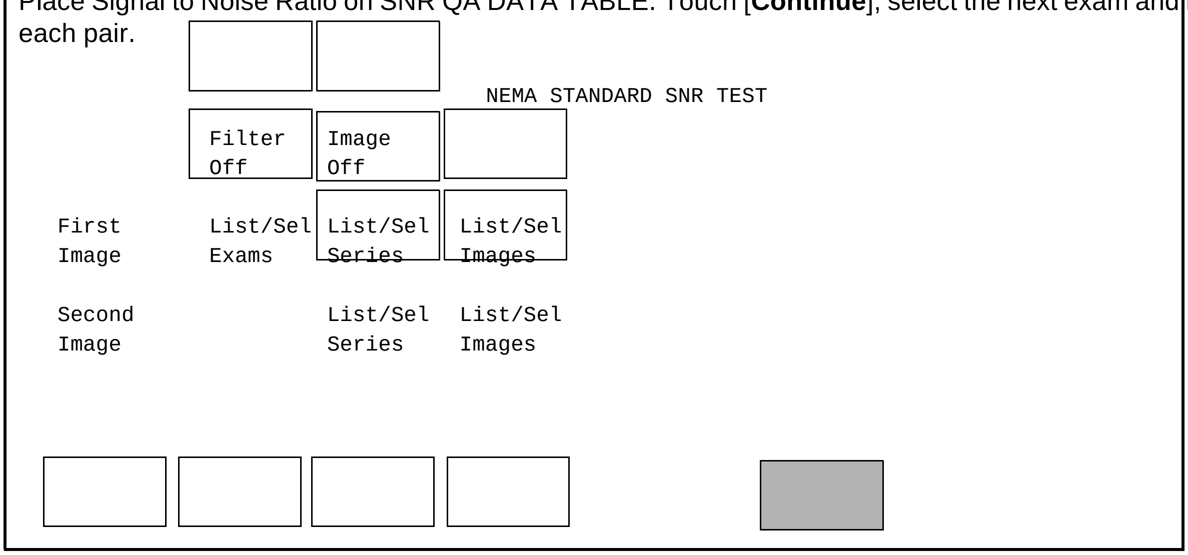
2. Enter first image exam, series, and image numbers. If exam, series, or image numbers are not known, select [**List Exams**], [**List Series**], or [**List Images**] to display list to choose from. Enter second image series and image number. The second image exam number is defaulted to the same as first image.

Note

Image number selection must be back lit (highlighted) to be able to enter information. Use Switch key on keyboard to transfer control from left to right side of Touch Screen.

3. Select [**Filter Off**] and [**Image Off**]. Touch [**Accept**] to begin analysis. The final values are displayed on the screen.

4. Place Signal to Noise Ratio on SNR QA DATA TABLE. Touch [**Continue**], select the next exam and repeat for each pair.



Note: Accept changes to continue only after an analysis has been performed.

SNR TEST SCREEN
ILLUSTRATION 3-4

```
Cancel      Utility  MR Tools  Image                      Accept
            Main     Main      Quality
```

3-4 CHECKING THE PROTECTION PIN DIODES WITH DIGITAL MULTIMETER (DMM)

1. Select the **DIODE TEST** function on the Digital Multimeter (DMM).

2. Connect the **NEGATIVE** lead of the DMM to the external connector ground pin. Connect the **POSITIVE** lead to the external cable connector signal pin. Refer to TABLE 3-2 and ILLUSTRATION 3-5 (Phased Array) or Table 3-3 and ILLUSTRATION 3-6 (Single-Receiver).

3. A reading of 0.400 to 0.900 should be observed on the DMM.

4. If a reading below 0.400 is observed, either the output cable is shorted or the PIN diode is shorted.

5. If a reading above 0.900 is observed in step 2, the PIN diode or the cable is open.

6. Connect the **NEGATIVE** lead of the DMM to the external cable connector signal pin. Connect the **POSITIVE** lead to the external cable connector ground pin. Refer to TABLE 3-2 and ILLUSTRATION 3-5 (Phased Array) or Table 3-3 and ILLUSTRATION 3-6 (Single-Receiver).

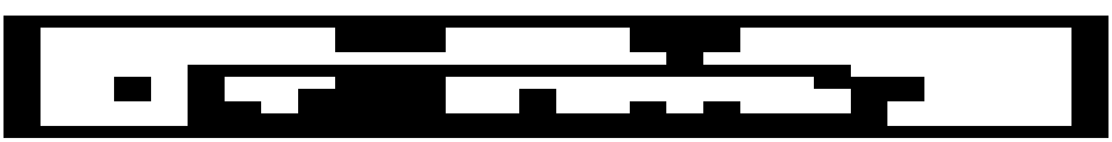
7. A reading of **INFINITY** should be observed on the DMM.

8. If a reading below **INFINITY** is observed, either the output cable is defective or the PIN diode is shorted or leaky.

9. If any of the above conditions fail, check the PIN diodes directly at the coils, and replace as necessary. Refer to Section 4-3, REPLACING THE PROTECTION/DECOUPLING PIN DIODES.

TABLE 3-2
DIODE TEST CONNECTIONS - PHASED ARRAY COIL

	POSITIVE LEAD CONNECTION		**NEGATIVE LEAD CONNECTION**	
COIL NUMBER	**FOR STEP 2**	**FOR STEP 6**	**FOR STEP 2**	**FOR STEP 6**



PIN LOCATIONS FOR PHASED ARRAY CONNECTOR
ILLUSTRATION 3-5

1 (right)	A3/B3	A2/B2	A2/B2	A3/B3
2 (left)	A9/B9	A2/B2	A2/B2	A9/B9

3-5 CHECKING THE EXTERNAL CABLE

1. Select the **DIODE TEST** function on the Digital Multimeter (DMM).

2. Connect the **NEGATIVE** lead of the DMM to the external cable connector ground pin. Connect the **POSITIVE** lead to the external cable connector signal pin. Refer to TABLE 3-2 and ILLUSTRATION 3-5 (Phased Array) or Table3-3 and ILLUSTRATION 3-6 (Single-Receiver).

3. Flex the external cable, especially near the connectors and the strain relief, and observe that a reading of 0.400 to 0.900 should remain on the DMM, with no instability of fluctuations.

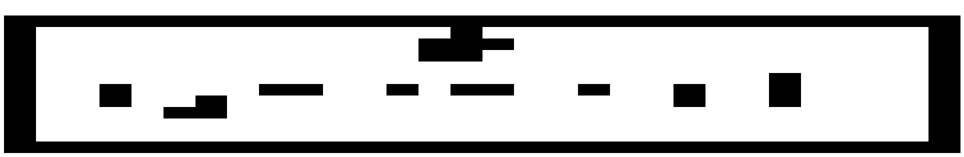
4. Connect the **NEGATIVE** lead of the DMM to the external cable connector signal pin. Connect the **POSITIVE** lead to the external cable connector ground pin. Refer to TABLE 3-2 and ILLUSTRATION 3-5 (Phased Array) or TABLE 3-3 and ILLUSTRATION 3-6 (Single-Receiver).

5. Flex the external cable, especially near the connectors and the strain relief, and observe that a reading of **INFINITY** should remain on the DMM, with no instability or fluctuations.

6. If the cable fails any of the above tests, it should be replaced, Refer to Section 4-2, REPLACING THE EXTERNAL CABLE.

TABLE 3-3
DIODE TEST CONNECTIONS -SINGLE RECEIVER

	POSITIVE LEAD CONNECTION		**NEGATIVE LEAD CONNECTION**	
COIL NUMBER	**FOR STEP 2**	**FOR STEP 6**	**FOR STEP 2**	**FOR STEP 6**
1 (right)	A2/B2	A1/B1	A1/B1	A2/B2
2 (left)	A9/B9	A1/B1	A1/B1	A9/B9



PIN LOCATIONS FOR SINGLE- RECEIVER CONNECTOR
ILLUSTRATION 3-6

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SECTION 4 - REPLACEMENT/MAINTENANCE

4-1 DISASSEMBLY/ASSEMBLY OF BREAST COIL

4-2 REPLACING THE EXTERNAL CABLE

4-3 REPLACING THE PROTECTION/DECOUPLING PIN DIODES

CAUTION

Use a 25W solder iron and protect the circuit board from static electricity. Be careful not to damage the circuit board by overheating.

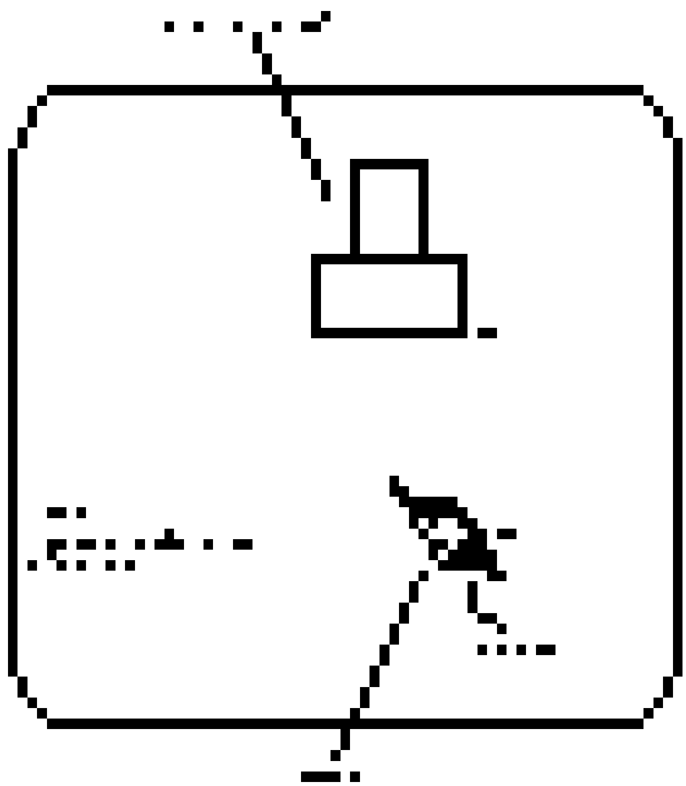
Note

Each antenna loop element is tuned at the factory to provide the proper input impedance. Circuit boards are not field replaceable. The PIN diodes are the only field replaceable component on the circuit boards.

1. Disassemble the top and bottom housings, if necessary. Refer to Section 4-1.

2. Use an anti-static wrist band when touching the circuit board.

3. Replace the PIN diode (D1) only with an exact replacement diode. OBSERVE POLARITY. See Renewal Parts. See ILLUSTRATION 4-1.



DIODE REPLACEMENT
ILLUSTRATION 4-1

4-4 CHECKING THE CABLES

1. Check the external cable for cracks or breaks once each week. Replace the external cable if any damage or wear is found. Refer to Section 4-2, REPLACING THE EXTERNAL CABLE, for replacement procedure.

4-5 CLEANING THE BREAST COIL

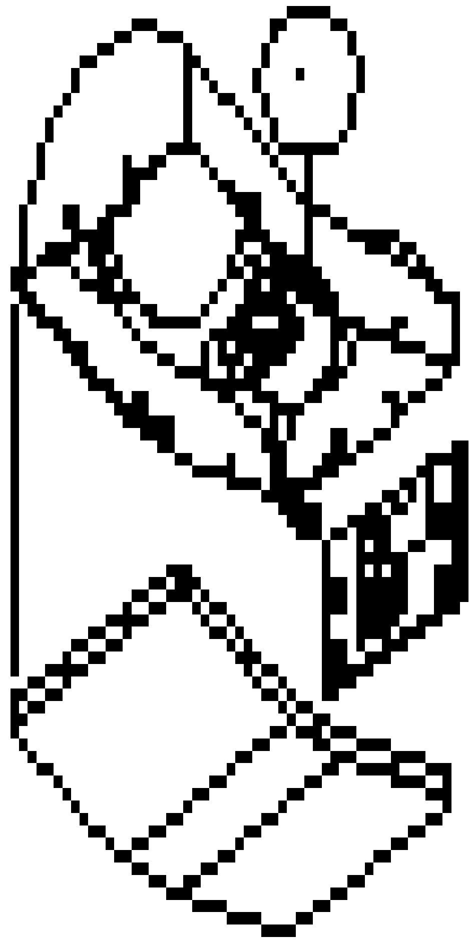
CAUTION

Avoid damaging sensitive electronic parts. Do not spray or pour dishwashing solution directly onto the Breast coil, or external cable. Do not use alcohol to clean the Breast coil. Never submerge the Breast coil in any liquid.

1. Clean the Breast coil and external cable with a mild dishwashing liquid and water solution. Wet a soft cloth with the solution and proceed to clean.

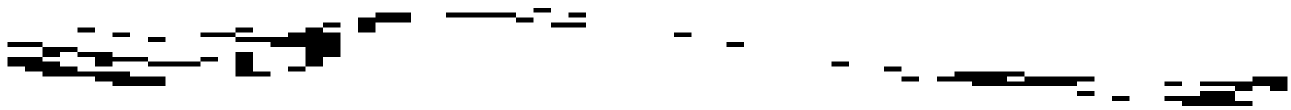
SECTION 5 - RENEWAL PARTS

BREAST COIL WITHOUT CABLE 46-328296P1



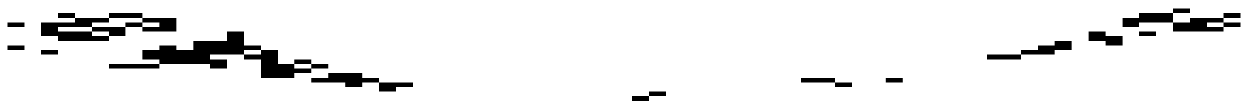
Item	Part Number	FRU	Name	Quantity	Description/Remarks
1	46-221735P1	2	Diode	1	UM9415 PIN Diode
2	46-328296P1	1	Breast Coil	1	Breast Coil without Cable
3	46-317586G1	2	Phantom	1	100 mm TLT Phantom

PHASED ARRAY CABLE REPAIR KIT 2100937-5



Item	Part Number	FRU	Name	Quantity	Description/Remarks
1	2100937-5	2	Cable Kit	1	Phased Array Cable Repair Kit

SINGLE-RECEIVER CABLE REPAIR KIT 2100937-6



Item	Part Number	FRU	Name	Quantity	Description/Remarks
1	2100937-6	2	Cable Kit	1	Single-Receiver Cable Repair Kit

Blank

SNR QA DATA TABLE

Use the space provided below to record the calculated signal to noise ratio—SNR, data obtained from Section 3, FUNCTIONAL CHECKS. After recording the SNR data obtained during the initial coil installation, record subsequent SNR data in column 2 — 4 (for single-receiver coil) or column 5 (for phased arrray coil) below and the date in column 1 as a periodic QA check. If the ratio of any of the coils found is not greater than 85%, then there is a problem in the coil or the MR system.

Original SNR data obtained at initial coil installation:

Body Coil SNR Value: ______________________

Breast SNR value: ____________Unilateral Left _____Both ________Unilateral Right _________

Date: ______________

SNR DATA QA (QUALITY ASSURANCE) CHECK:

1 Date	2 Unilateral Left QA SNR-Value:	3 Both QA SNR-Value:	4 Unilateral Right QA SNR-Value:	5 Phased Array Coil QA SNR-Value:	6 Is new value divided by original value >85%?
____	________	________	________	________	________ Y/N
____	________	________	________	________	________ Y/N
____	________	________	________	________	________ Y/N
____	________	________	________	________	________ Y/N
____	________	________	________	________	________ Y/N
____	________	________	________	________	________ Y/N
____	________	________	________	________	________ Y/N
____	________	________	________	________	________ Y/N
____	________	________	________	________	________ Y/N
____	________	________	________	________	________ Y/N
____	________	________	________	________	________ Y/N

Make additional copies of this document as needed.

DEFECTIVE SURFACE COIL RETURN FORM

NOTE

For proper assessment of defective returned coils, both sides of this form should be completely filled out and accompany all returned coils. Include films or prints of any image quality complaints with a description of the scan prescription used.

DATE: ____________________

SITE NAME: ____________________

SITE ADDRESS: ____________________

SERVICE ENGINEER: ____________________

COIL SERIAL NUMBER: ____________________

DATE COIL INSTALLED: ____________________

DESCRIPTION OF COIL PROBLEM: ____________________

(Please be descriptive; ”Broken” is not enough) ____________________

Note

An alternate proprietary procedure is available to GE use and to customers with a valid Advanced Service Package Limited License. Refer to “TLT PROCEDURE” located in Direction 15491, *Signa Advantage 1.5T, 1.0T & 0.5T System Troubleshooting Guide.* Use the table to record the TLT results of the defective coil.

TABLE DR-1
TLT DATA FOR DEFECTIVE SURFACE COIL

SITE:	**NAME:**	**DATE:**
TLT FILE NUMBER:		
SNR: NOISE		
SNR MEAN		
SNR SIGNAL		
SNR AREA		
TRMAP: 89–91	%	%
85–95	%	%
65–115	%	%
FLIP MEAN		
FLIP SDV		
RCV MEAN		
RCV SDV		
DIFF MEAN		
DIFF SDV		

TABLE DR-2
TLT DATA FOR DEFECTIVE PHASED ARRAY COIL

SITE:	**NAME:**	**DATE:**
TLT FILE NUMBER:		
SNR: NOISE (0)		
NOISE (1)		
NOISE (2)		
NOISE (3)		
NOISE (C)		
SNR MEAN (0)		
SNR MEAN (1)		
SNR MEAN (2)		
SNR MEAN (3)		
SNR MEAN (C)		
TRMAP: 89–91 (C)	%	%
85–95 (C)	%	%
65–115 (C)	%	%
FLIP MEAN (C)		
FLIP SDV (C)		
RCV MEAN (C)		
RCV SDV (C)		
DIFF MEAN (C)		
DIFF SDV (C)		

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